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**Division: A2**

### **ASSIGNMENT No: 3**

**TITLE:** Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

**Problem Statement:** Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

**Learning Objectives:** To implement method overloading and use static variables and methods in Java programs.

**Theory:** Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

#### **Exercise**

**1.**Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

**Program:**

```
C: > Users > Ayush > Downloads > class-Calculator.java > Java > Calc > add(double a, double b)

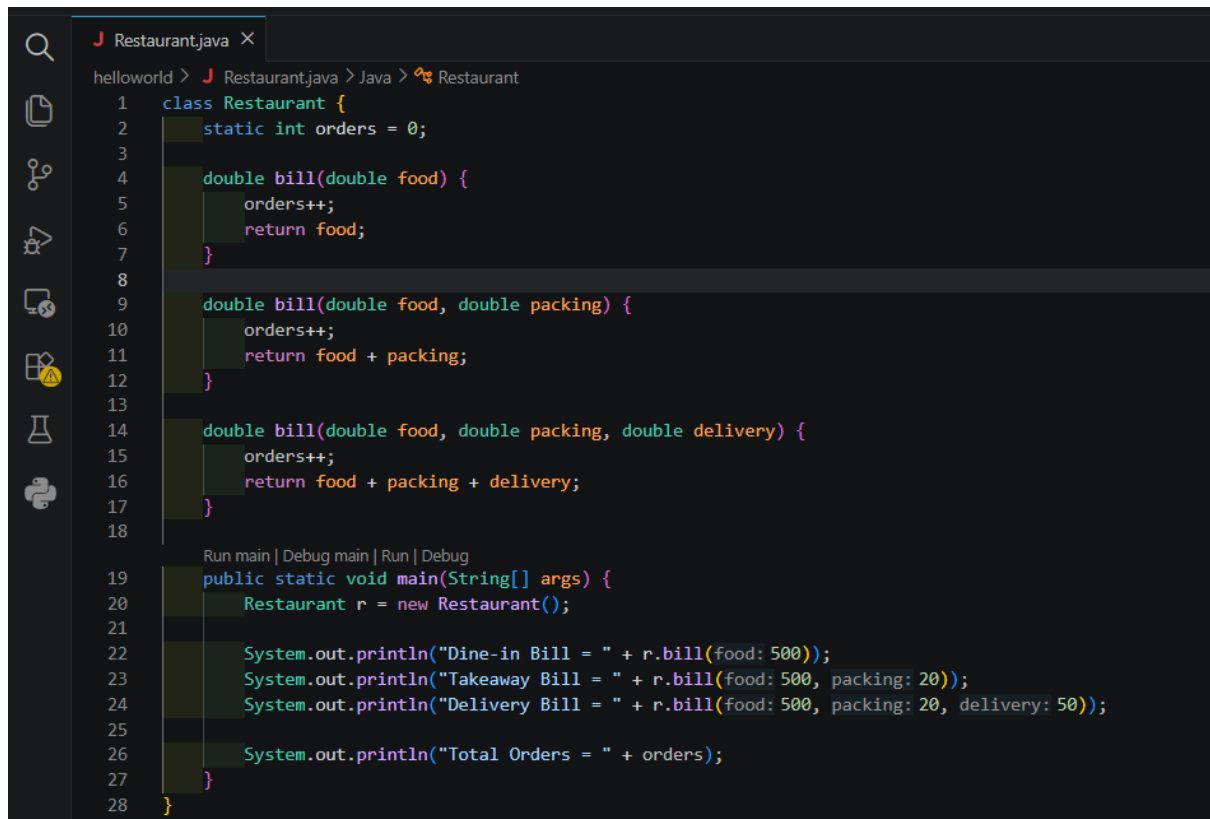
1  class Calc {
2      static int count = 0;
3
4      int add(int a, int b) {
5          count++;
6          return a + b;
7      }
8
9      double add(double a, double b) {
10         count++;
11         return a + b;
12     }
13
14     public static void main(String[] args) {
15         Calc c = new Calc();
16
17         System.out.println("Integer Sum = " + c.add(a: 10, b: 20));
18         System.out.println("Decimal Sum = " + c.add(a: 10.5, b: 20.5));
19
20         System.out.println("Total Calculations = " + count);
21     }
22 }
```

Output:

```
PS C:\Users\Ayush\projects> & 'C:\Us
eStorage\44eea058444828944518bac73083
Integer Sum = 30
Decimal Sum = 31.0
Total Calculations = 2
PS C:\Users\Ayush\projects>
```

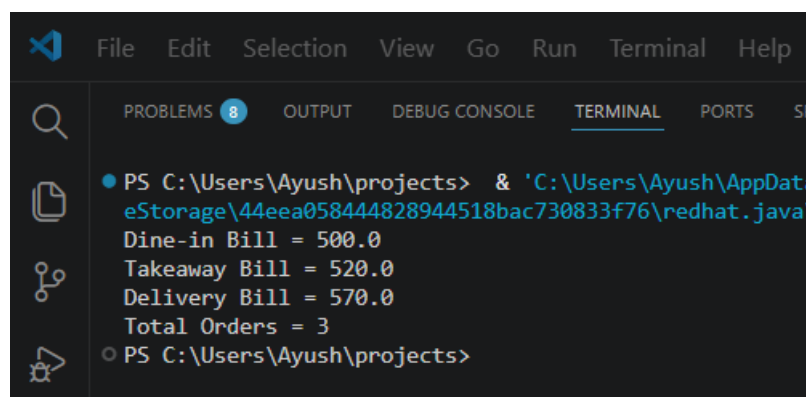
2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

## Program:



```
1 class Restaurant {
2     static int orders = 0;
3
4     double bill(double food) {
5         orders++;
6         return food;
7     }
8
9     double bill(double food, double packing) {
10        orders++;
11        return food + packing;
12    }
13
14    double bill(double food, double packing, double delivery) {
15        orders++;
16        return food + packing + delivery;
17    }
18
19    Run main | Debug main | Run | Debug
20    public static void main(String[] args) {
21        Restaurant r = new Restaurant();
22
23        System.out.println("Dine-in Bill = " + r.bill(food: 500));
24        System.out.println("Takeaway Bill = " + r.bill(food: 500, packing: 20));
25        System.out.println("Delivery Bill = " + r.bill(food: 500, packing: 20, delivery: 50));
26
27        System.out.println("Total Orders = " + orders);
28    }
29 }
```

## Output:



```
PS C:\Users\Ayush\projects> java -cp 'C:\Users\Ayush\AppData\Local\Temp\44eea058444828944518bac730833f76\redhat.java' Restaurant
Dine-in Bill = 500.0
Takeaway Bill = 520.0
Delivery Bill = 570.0
Total Orders = 3
PS C:\Users\Ayush\projects>
```